11.5.4 Functions of Water

Major functions of water :

1) **Carrier of nutrients:** Every nutrient in soluble form in water is carried from intestines to tissues through blood.

2) **Constituent of body fluids:** Water is the major constituent of all body fluids such as blood, urine, sweat, lymph.

3) **Regulation of body temperature:** Water helps to regulate and control body temperature. Heat is produced when food is burnt for energy. Water is evaporated through respiration and sweat and body temperature is maintained normal. Body's heat is lost through the skin, lungs, urine and faeces.

4) **Protection of delicate organs:** Water found around lungs, heart and brain protects them from outer injury.

5) **Water as lubricant:** Water acts as a lubricant in joints. Water around joints helps in normal movement.
It is an essential constituent of all the cells of the body and the internal environment.



11.5.5 Requirements

Requirement of water varies with climate, dietary constituents, activities and surface area of the body. As a rule a person should take enough water to excrete about 1200 –1500 ml of urine per day. In tropics because of greater water loss through perspiration, increased water intake is required to maintain urine volume. Normal intake of water ranges between 8 – 10 glasses per day.

Water is lost through feces, urine, lungs (expiration) skin (invisible perspiration and visible perspiration) amounting to about 2-3 liters per day. During infections and fever, the liquid intake should be increased as losses are higher. A moderate amount of water taken with or preceding a meal aids in digestion.

11.5.6 Dehydration

Dehydration results in extreme deficiency of water and fluids. Symptoms of dehydration are fatigue, headache, sullenness and in extreme cases, collapse.

The steps in the progression of dehydration are as follows.

1. Thirst
2. Decreasing blood volume, impaired physical performance.
3. Increased effort for physical work, nausea.
4. Failure to regulate excess temperature.
5. Muscle spasms.
6. Failing renal function, less or no urine formed.

Excessive loss of water takes place due to vomiting, diarrohea, haemorrhage, excessive perspiration, exudating, burns, uncontrolled diabetes mellitus, fever and hot weather. It can be fatal and can be easily avoided by proper fluid intake or oral rehydration therapy.

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